

NITRO-D: Native Integer-Only Training of Deep Convolutional Neural Networks

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Abstract. Quantization is a pivotal technique for managing the growing computational and memory demands of Deep Neural Networks (DNNs). By reducing the number of bits used to represent weights and activations (typically from 32-bit Floating-Point (FP) to 16-bit or 8-bit integers), quantization reduces memory footprint, energy consumption, and execution time of DNNs. However, most existing methods typically target DNN inference, while training still relies on FP operations, limiting applicability in environments where FP arithmetic is unavailable. To date, only one prior work has addressed integer-only training, and only for Multi-Layer Perceptron (MLP) architectures. This paper introduces NITRO-D, a novel framework for training deep integer-only Convolutional Neural Networks (CNNs) that operate entirely in the integer domain for both training and inference. NITRO-D enables training of integer CNNs without requiring a separate quantization scheme. Specifically, it introduces a novel architecture that integrates multiple *local-loss blocks*, which include the proposed *NITRO-Scaling* layer and *NITRO-ReLU* activation function. The proposed framework also features a novel learning algorithm that employs local error signals and leverages *IntegerSGD*, an optimizer specifically designed for integer computations. NITRO-D is implemented as an open-source Python library. Extensive evaluations on state-of-the-art image recognition datasets demonstrate its effectiveness. For integer-only MLPs, NITRO-D improves test accuracy by up to +5.96% over the state-of-the-art. It also successfully trains integer-only CNNs, reducing memory requirements and energy consumption by up to 76.14% and 32.42%, respectively, compared to the traditional FP backpropagation algorithm.

1 Introduction

In recent years, quantization of Deep Neural Networks (DNNs) has emerged as a pivotal trend in both the scientific and technological domains [11]. This technique aims at lowering the computational demand and memory requirements of DNNs by reducing the number of bits to represent weights and activations (e.g., from 32-bit Floating-Point (FP) to 16-bit or 8-bit integers). However, existing quantization frameworks for DNNs retain a dependence on FP operations during training. In this perspective, the aim of this paper is to address the following research question: *Is it possible to train arbitrarily deep Convolutional Neural Networks (CNNs) using only integer values for both training and inference?*

The literature in this field is extensive but highly fragmented. Existing frameworks for integer-only DNNs can be grouped into three main categories. The first, *quantization-aware inference*, applies quantization only at inference time, while training relies on FP operations to compute gradients [18, 16, 7, 29, 44, 2, 22, 35, 24, 23]. The second, *quantization-aware training*, extends quantization to the training process, further exploiting the advantages of integer arithmetic at the cost of additional complexity in the quantization scheme [43, 40, 41, 38, 5, 17, 26, 42, 4, 33, 10]. Despite these advancements, such approaches still partially rely on FP arithmetic or employ custom fixed-point numeric formats that emulate it. The last category, *native integer-only training*, uses integer-only arithmetic during both inference and training. To date, the only solution in this category targets Multi-Layer Perceptron (MLP) architectures [32].

This paper introduces NITRO-D, a native integer-only framework for training deep CNNs exclusively employing integer arithmetic. This approach eliminates the need for any explicit quantization steps or FP operations during training. NITRO-D features a novel network architecture composed of multiple local-loss blocks and introduces novel components, such as the *NITRO-Scaling* layer and the *NITRO-ReLU* activation function, which are specifically designed to prevent integer overflow. In addition, NITRO-D employs a novel learning algorithm adapted from Local Error Signals (LES) [27] to enable integer-only training. Specifically, this algorithm relies on two key innovations: the *IntegerSGD* optimizer and the *NITRO Amplification Factor*, which is used to calibrate the learning rate. To the best of our knowledge, NITRO-D is the first framework in the literature to enable native integer-only training of deep CNNs. Thanks to the simplicity of integer arithmetic [8], NITRO-D offers reduced hardware complexity and lower energy consumption. Integer operations can be executed using smaller bit-widths (such as 8-bit and 16-bit), enabling faster execution times and lower memory requirements compared to traditional 32-bit FP operations. NITRO-D is released as an open-source library to the scientific community¹.

The main contributions of this work are:

1. The NITRO-D architecture, which enables the construction of arbitrarily deep CNNs operating entirely in the integer domain.
2. The NITRO-D learning algorithm, which enables native integer-only training of the NITRO-D architecture without requiring neither quantization schemes nor FP operations.

Experimental results on multi-class image classification benchmarks show test accuracy improvements of up to +5.96% for integer-

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¹ <https://github.com/AI-Tech-Research-Lab/nitro-d>